# FUNDAMENTAL THEOREM OF FINITE GALOIS THEORY: SUBGROUPS CORRESPOND TO SUBEXTENSIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: Both squeeze ingredients are the two lemmas proved inside the span's own first paragraph: |Gal(L/M)| = [L : M] for subextensions and [L : L^H] = |H| for subgroups.

*Editorial note.* The label theorem-galois-theory names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own theorem environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Context spans: Gal(E/F) notation and the fixed field L^H
- Same chapter: lemma-galois-goes-up (L/M is Galois for every subextension L/M/K of a Galois L/K), lemma-finite-goes-up (finiteness goes up), lemma-finite-Galois (finite extension is Galois iff |Aut| = degree), lemma-galois-over-fixed-field ([L : L^H] = |H| for finite H), lemma-lift-maps (restriction of automorphisms to subextensions)

*Source reference.* Original reference 09I1: sources/stacks-fields/pinned-original-source.tex, source lines 2881–2881. Prerequisite pointer invoked at 3019; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09G5: sources/stacks-fields/pinned-original-source.tex, source lines 668–668. Prerequisite pointer invoked at 3018; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09I2: sources/stacks-fields/pinned-original-source.tex, source lines 2911–2911. Prerequisite pointer invoked at 3016; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09I3: sources/stacks-fields/pinned-original-source.tex, source lines 2944–2944. Prerequisite pointer invoked at 3021; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0BME: sources/stacks-fields/pinned-original-source.tex, source lines 2163–2163. Prerequisite pointer invoked at 3034; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Definition 0.1.** If $E/F$ is a Galois extension, then the group $\mathrm{Aut}(E/F)$ is called the *Galois group* and it is denoted $\mathrm{Gal}(E/F)$.

Let $G$ be a group acting on a field $K$ (by field automorphisms). We will often use the notation

$$K^G = \{x \in K \mid \sigma(x) = x \ \forall \sigma \in G\}$$

and we will call this the *fixed field* for the action of $G$ on $K$.

**Theorem 0.2** (Fundamental theorem of Galois theory)**.** *Let $L/K$ be a finite Galois extension with Galois group $G$. Then we have $K = L^G$ and the map*

$$\{\textit{subgroups of } G\} \longrightarrow \{\textit{subextensions } L/M/K\}, \quad H \longmapsto L^H$$

*is a bijection whose inverse maps $M$ to $Gal(L/M)$. The normal subgroups $H$ of $G$ correspond exactly to those subextensions $M$ with $M/K$ Galois.*

*Proof.* By Lemma 09I2 given a subextension $L/M/K$ the extension $L/M$ is Galois. Of course $L/M$ is also finite (Lemma 09G5). Thus $|\mathrm{Gal}(L/M)| = [L : M]$ by Lemma 09I1. Conversely, if $H \subset G$ is a finite subgroup, then $[L : L^H] = |H|$ by Lemma 09I3. It follows formally from these two observations that we obtain a bijective correspondence as in the theorem.

If $H \subset G$ is normal, then $L^H$ is fixed by the action of $G$ and we obtain a canonical map $G/H \to \mathrm{Aut}(L^H/K)$. This map has to be injective as $\mathrm{Gal}(L/L^H) = H$. Hence $|G/H| = [L^H : K]$ and $L^H$ is Galois by Lemma 09I1.

Conversely, assume that $K \subset M \subset L$ with $M/K$ Galois. By Lemma 0BME we see that every element $\tau \in \mathrm{Gal}(L/K)$ induces an element $\tau|_M \in \mathrm{Gal}(M/K)$. This induces a homomorphism of Galois groups $\mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$ whose kernel is $H$. Thus $H$ is a normal subgroup. □